Predictability Effects on Memory Retrieval Are Structure-Dependent

Summer Fan Xia & Ming Xiang, The University of Chicago | summerxia@uchicago.edu

Predictable information is remembered more accurately and places reduced demands on working memory (WM) [1-2]. This is possibly because high-probability input has more robust long-term memory (LTM) representations, which can facilitate more efficient memory retrieval [3]. At the same time, however, it also has been shown that retrieving individual elements embedded within a high-probability unit could be costly, due to the fact that chunking high-probability information into a representational unit might lead to compression and recoding of the input, rendering chunk-internal information less accessible, and a costly decoding process would be required for retrieving chunk-internal elements [4-5]. Using ellipsis structure as a case study, the current study found support for both mechanisms. But crucially, whether we observe high-probability advantage or disadvantage is modulated by the specific structures involved, in particular, whether it is about a subject-predicate relation or a verb-object relation.

Methods We used a bidirectional self-paced reading paradigm [6]. **[EXP1]** (N=123, 24 items) manipulated the probability of VPs ("wield a sword/pan") by varying the object NP's surprisal given its verb. We tested two ellipsis structures that produced different RETRIEVAL TYPEs: NP ellipsis (NPE) required *Elemental Retrieval* of individual elements ("sword/pan") from a larger high/low-probability VP ("wield a sword/pan"), while VP ellipsis (VPE) involved *Holistic Retrieval* of the entire VP. **[EXP2]** (N=123, 24 items) included only VPE structures. We manipulated probability either at the VP level ("wield a sword/pan", as in EXP1) or at the subject-predicate level ("The knight/runner wielded the sword"), where the entire VP's surprisal was calculated conditional on the subject NP. This manipulation also created *Holistic* and *Elemental Retrieval* conditions: in *Elemental Retrieval*, readers retrieved VPs ("wielded a sword") embedded within larger units ("The knight wielded the sword"). In both experiments, each participant saw only one condition per item and answered a comprehension question after each trial. We used linear mixed-effects models to analyze first-pass reading times (FPRT) and proportions of first-pass regressions (RegProp) at the retrieval site ("did too" and "one") and the first spillover region.

Results [EXP1] For FPRT, the mixed-effects model found a main effect of probability ($p<.05$) at the first spillover region; RegProp showed an interaction between probability and RETRIEVAL TYPE at the retrieval site ($p<.05$) and the first spillover ($p<.001$) region. Follow-up analyses revealed that the interaction reflected different loci of probability effects across retrieval modes: in *Holistic Retrieval*, probability affected FPRT ($p<.05$) and RegProp ($p=.06$) at the retrieval region; in *Elemental Retrieval*, probability affected FPRT ($p<.05$) and RegProp ($p<.001$) at the first spillover region. Across both RETRIEVAL TYPEs, readers showed shorter FPRTs and were less likely to regress when the VP was highly probable and thus more likely to form a memory unit. **[EXP2]** RegProp again showed a significant interaction between probability and RETRIEVAL TYPE at the first spillover region ($p<.001$). Follow-up analyses showed that *Holistic Retrieval* replicated the patterns found in EXP1: probability affected FPRT ($p<.05$) at the retrieval site, and RegProp at the retrieval site ($p=.06$) and the first spillover region ($p<.001$). In *Elemental Retrieval*, probability also affected RegProp ($p<.001$) at the first spillover region. However, the direction of the probability effect reversed: high-probability subject-predicate combinations elicited more regressions, suggesting increased retrieval cost.

Conclusions The results of *Holistic Retrieval* confirmed better memory for predictable input. *Elemental Retrieval* revealed more nuanced patterns that varied across verb-object and subject-predicate relations: retrieving object NPs from VPs showed a probability advantage, whereas retrieving VPs embedded within a subject-predicate relation showed a probability disadvantage. These findings suggest that syntactic structure shapes memory strategy. Verb-object relations are known to be more tightly linked than subject-predicate relations [7]. During online processing, individuals may exploit robust, pre-existing LTM representations of verb-object relations to facilitate memory retrieval. By contrast, the generally looser association between a subject and its predicate may trigger compression or recoding-based chunking to bind them together, which in turn renders a decoding cost when the predicate needs to be retrieved.

Table 1. Stimuli. Curly brackets indicate the hypothetical memory units. Underlines mark the information that must be retrieved at the ellipsis site.

EXP1		
RETRIEVAL TYPE	PROBABILITY	
Holistic	High:	Louis {wielded a sword}, and simultaneously Kyle *did too* with visible effort.
	Low:	Louis wielded a pan, and simultaneously Kyle *did too* with visible effort.
Elemental	High:	Louis {wielded a sword}, and suddenly Kyle raised *one* as well with visible effort.
	Low:	Louis wielded a pan, and suddenly Kyle raised *one* as well with visible effort.

EXP2		
RETRIEVAL TYPE	PROBABILITY	
Holistic	High:	Louis {wielded a sword}, and simultaneously Kyle *did too* with visible effort.
	Low:	Louis wielded a pan, and simultaneously Kyle *did too* with visible effort.
Elementary	High:	{The knight wielded a sword}, and simultaneously the mentor *did too* with visible effort.
	Low:	The runner {wielded a sword}, and simultaneously the mentor *did too* with visible effort.

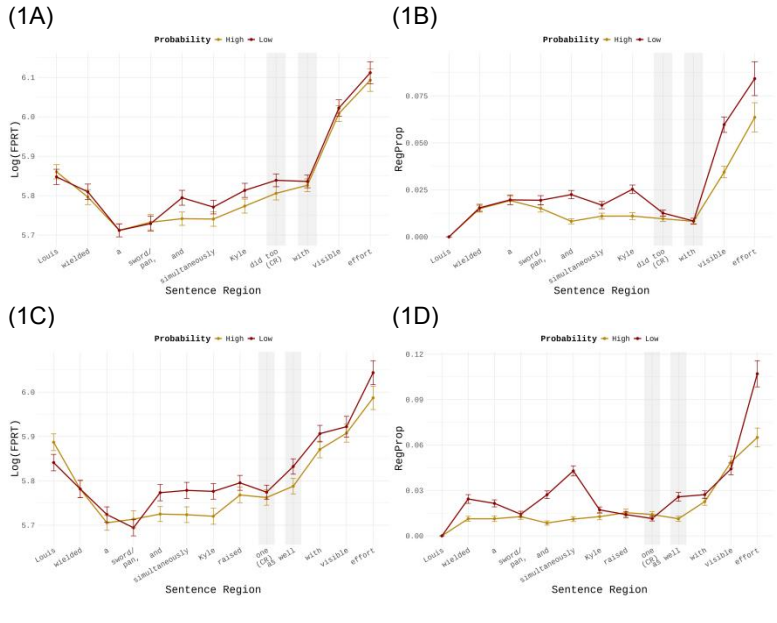

Figure 1. **EXP1 Results.** Upper and lower panels illustrate reading patterns for *Holistic Retrieval* and *Elemental Retrieval*, respectively. (1A) & (1C) display the probability effects on FPRTs across all sentence regions. (1B) & (1D) present the probability effects on RegProp. The retrieval site and the first spillover regions are shaded in gray.

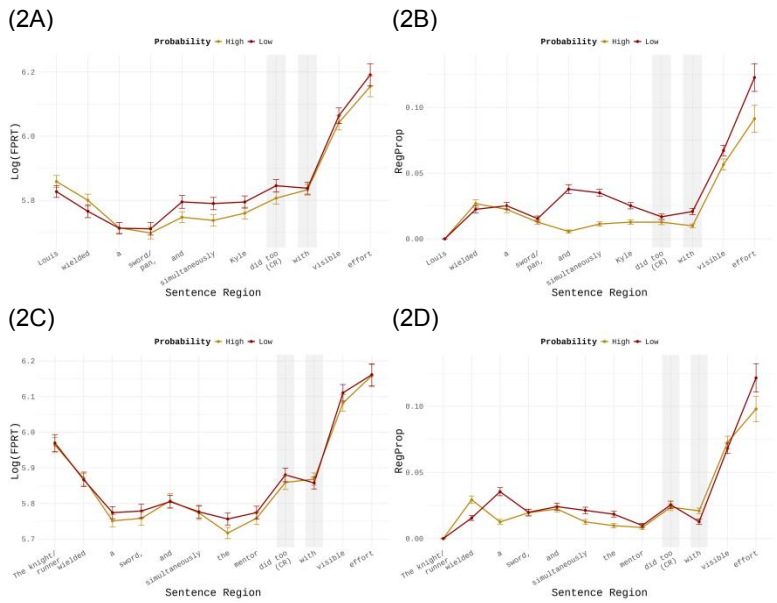

Figure 2. **EXP2 Results.** Format mirrors Figure 1. Upper and lower panels present *Holistic* and *Elemental Retrieval*, respectively. Panels (2A) & (2C) show FPRTs; Panels (2B) & 2D) present RegProp patterns.

References: [1] Thalmann et al. (2019). *J. Exp. Psychol. LMC*. [2] Brady et al. (2009). *J. Exp. Psychol. Gen*. [3] Norris et al. (2020). *J. Exp. Psychol. LMC*. [4] Cowan et al. (2004). *Psychological Science*. [5] Huang & Awh (2018). *Scientific Reports*. [6] Paape & Vasishth (2022). *Cognitive Science*. [7] Rooryck & Zaring (1996). *Phrase Structure and the Lexicon*.